

Dosimetric Characterization of a Novel Silicon Carbide p-n Junction Detector under Clinical Photon Beams

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Version typeset March 19, 2026

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Abstract

Background: Wide-bandgap semiconductors such as silicon carbide (SiC) offer ultra-low leakage current and high radiation hardness, making them attractive for real-time dosimetry in clinical megavoltage photon beams.

Purpose: This work presents a dosimetric characterization of a novel 4H-SiC planar p-n junction detector for radiotherapy, aiming to assess its performance as an active dosimeter in clinical MV photon beams.

Methods: The detector was investigated under 6 MV, 10 MV, and 15 MV photon beams from a clinical linear accelerator. Pre-irradiation electrical characterization (I-V and C-V) was performed to quantify leakage current and depletion behavior. For dosimetric tests, the device was operated at 30 V reverse bias and read out with a Keithley 6517B electrometer through low-noise triaxial cabling. Measurements were carried out in a water phantom under reference geometry (SSD = 100 cm, $10 \times 10 \text{ cm}^2$ field). A Farmer-type ionization chamber calibrated in terms of absorbed dose to water provided the traceable reference for cross-calibration and comparisons. The detector response was evaluated in terms of dose linearity, dose-rate dependence, energy dependence across beam qualities, percent depth-dose (PDD) curves, angular response (polar and azimuthal scans), and temperature dependence.

Results: The SiC diode showed rectifying behavior with picoampere-level reverse leakage at 30 V, and C-V data indicated full depletion at $\sim 10 \text{ V}$, supporting efficient charge collection in over-depleted operation. Under irradiation, the collected charge was linear with delivered dose (maximum deviation $< 1\%$, $R^2 > 0.99$), with a sensitivity of 92.5 nC Gy^{-1} at 6 MV. The response remained within $\pm 1\%$ over the investigated dose-rate range and varied by less than $\sim 2\%$ between 6–15 MV after normalization. PDD curves measured with the SiC detector closely reproduced the reference distributions: d_{max} agreed within 1 mm and differences beyond d_{max} were within 1%, while a small $\sim 2\%$ over-response was observed in the build-up region. The polar angular dependence remained within $\sim 3\%$, whereas a pronounced azimuthal modulation was

observed, attributable to the asymmetric orientation of the planar diode. Temperature tests showed a small negative trend corresponding to an average coefficient of about $-0.2\% \text{ } ^\circ\text{C}^{-1}$ between 20 and 40°C.

Conclusions: The proposed 4H-SiC p-n junction detector exhibits high signal-to-noise, excellent dose linearity, weak dose-rate and energy dependence, and accurate depth-dose measurements in clinical MV photon beams. Orientation control is required due to the observed azimuthal modulation, but overall the results support SiC as a robust solid-state dosimeter for clinical QA and beam characterization.

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1. Introduction

The increasing complexity of radiotherapy technique with photon beams, including intensity-modulated radiation therapy (IMRT), volumetric-modulated arc therapy (VMAT), and stereotactic approach, has made detector performance a central limiting factor in patient-specific quality assurance and beam data verification [1,2,3]. These deliveries combine steep dose gradients, small and dynamically varying apertures, and high monitor-unit rates, so the ideal dosimeter must provide high spatial resolution and real-time readout while preserving water-equivalent response, linearity over a wide dynamic range, minimal dependence on beam quality and dose rate, angular robustness, thermal stability, and long-term reproducibility [4,5]. In this framework, solid-state detectors have become increasingly attractive because they directly convert deposited energy into charge carriers within a micrometric-to-millimetric sensitive region, enabling compact geometries and fast signal collection. Compared to gas-filled chambers, the absence of an air cavity and the possibility of very small sensitive volumes reduce volume-averaging effects and facilitate measurements in high-gradient regions and small fields. Moreover, solid-state sensors can be integrated into arrays or multilayer concepts, enabling high-resolution mapping and potentially online dosimetry [6]. However, achieving metrologically reliable solid-state dosimetry in clinical MV photon beams requires careful control of material properties, device architecture, encapsulation, and readout, since perturbation effects and charge-collection phenomena can introduce energy, dose-rate, and directional dependencies [3,4]. Conventional detector families each address only a subset of these requirements. Thimble and plane-parallel ionization chambers remain the reference for absolute dose to water, yet their finite cavity size and effective point-of-measurement considerations limit spatial resolution and can bias small-field measurements. Passive systems such as thermoluminescent dosimeters (TLDs) provide small dimensions and integral dose, but they lack real-time readout and are affected by handling and readout variability, as well as potential fading and non-ideal response outside calibration conditions. Among solid-state options, silicon diodes offer high sensitivity and immediate readout but may exhibit over-response to low-energy scattered components, temperature sensitivity, and radiation-induced drifts that motivate pre-irradiation and periodic recalibration [7,8,9]. Synthetic diamond detectors mitigate several of these limitations through near tissue equivalence and excellent radiation hardness, but device cost and availability can constrain widespread adoption, and

detector-to-detector variability or stabilization procedures have historically required careful
characterization ^{DeAngelis2002_diamondDetector, Ciancaglion2012_diamondDetector, Laub2014_commercialDiamond}
^{10,11,12,13}. In recent years, wide-bandgap semiconductors have emerged as
a particularly promising solid-state platform to bridge the gap between the maturity and
ease of use of semiconductor dosimetry and the robustness demanded by modern clinical
workflows. In this context, silicon carbide (SiC) provides a compelling compromise between
established silicon processing and the intrinsic stability typical of wide-bandgap materials.
SiC combines a large bandgap (e.g., ~ 3.26 eV for 4H-SiC) with high breakdown field, en-
abling operation under reverse bias with ultra-low leakage currents and correspondingly low
electronic noise ^{Nava2008_SiCDetector, DeNapoli2022_SiCReview, LaVia2014_SiCOverview, Capan2022_SiCReview}
^{14,15,16,17}. Its strong Si-C bonds yield a high displacement energy threshold,
supporting radiation hardness and limiting sensitivity drifts over accumulated dose. From a
dosimetric perspective, while SiC is not as tissue-equivalent as diamond, its effective atomic
number is closer to water than silicon, and properly engineered device geometries can mini-
mize perturbations and support a weak dependence on MV photon beam quality. In addition,
high carrier saturation velocities and efficient depletion in thin epitaxial layers enable fast,
near-unity charge collection under clinically relevant irradiation conditions ^{Petringa2020_FirstSiC, Petringa}
^{18,19,20}. Building
on this solid-state perspective, the present work introduces a silicon carbide p-n junction
diode specifically designed for radiotherapy dose measurements and reports its dosimetric
characterization in a water phantom under clinical MV photon beams. We evaluate the
detector response in terms of sensitivity and dose linearity, dose-rate and energy dependence
across 6–15 MV, percent depth dose (PDD) measurements, angular response, temperature
dependence, and signal stability. Comparisons against established reference dosimetry are
used to assess water-equivalence and measurement accuracy under realistic clinical condi-
tions. By establishing a rigorous baseline for SiC p-n junction operation in MV photon
beams, this study aims to support the integration of SiC-based solid-state dosimeters into
routine clinical dosimetry and small-field/high-gradient QA workflows.

II. Methods

II.A. Device's Design

The adopted detector is based on a next-generation solid-state sensor manufactured in
4H-Silicon Carbide (4H-SiC). The device implements a planar p-n junction specifically

engineered for dosimetric measurements. The sensitive region is a $10\ \mu\text{m}$, lightly doped n-type epitaxial film (donor concentration $\sim 0.5\text{--}1 \times 10^{14}\ \text{cm}^{-3}$) grown on a highly doped p⁺ substrate ($N_A \approx 10^{19}\ \text{cm}^{-3}$). This layout yields a uniform high electric field throughout the active layer under reverse bias, enabling near-unity charge collection. Pre-irradiation electrical tests indicated full depletion at about 10 V, with stable operation under over-depleted conditions up to 30 V. Biasing in this range ensures saturated carrier drift and excellent charge-collection efficiency, including for fast or pulsed irradiations. The detector's active footprint is $4\ \text{mm}^2$ ($2 \times 2\ \text{mm}^2$), which is advantageous for clinical dosimetry where lateral field flatness and spatial resolution are critical. The chip is mounted on a custom circular printed circuit board (PCB) ^{Petringa2025,Petringa2026_Field}. For measurements under water-equivalent conditions, the assembly is placed inside a cylindrical metal holder and encapsulated in black epoxy, allowing safe immersion in a water phantom ^{Guarrera2025UHDR}. This packaging provides mechanical robustness while keeping the perturbation to the radiation field minimal (see Fig. ^{fig:Detector} [I](#)).

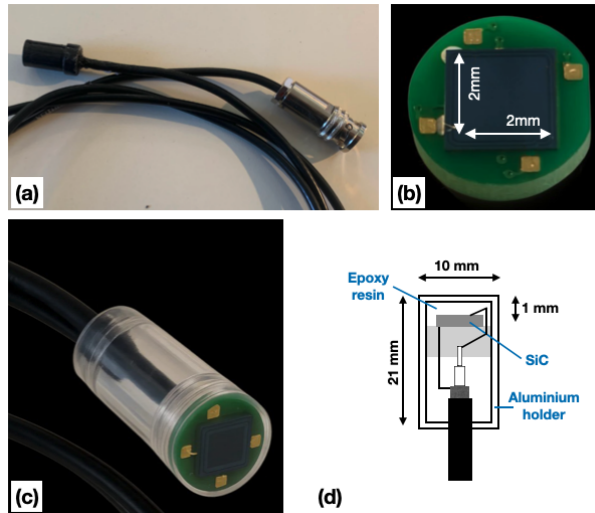


Figure 1: (a) Photograph of the finalized SiC dosimeter with its integral shielded cable and connector. (b) Technical top-view drawing of the packaged device mounted on the circular PCB; the sensitive area is $2 \times 2\ \text{mm}^2$ as indicated. (c) Technical rendering of the encapsulated assembly for water-phantom use, showing the cylindrical housing and the encapsulation. (d) Dimensioned cross-sectional drawing of the mechanical package: the SiC chip is seated in an aluminium holder and overmoulded with epoxy resin. The entrance epoxy window is $\sim 1\ \text{mm}$ thick; the outer envelope is $\sim 10\ \text{mm}$ in diameter and $\sim 21\ \text{mm}$ in length. The signal is routed through an insulated feedthrough to the external triaxial cable.

fig:Detector

Electrical connections to the diode are made via wire bonds, providing low-impedance paths to the readout. Thanks to the wide bandgap of SiC ($\approx 3.26\ \text{eV}$) and the high crystalline

quality of the epitaxial layers, the device exhibits extremely small reverse-bias leakage currents. This enables precise current readout in the picoampere range, essential for accurate dosimetry at low to moderate dose rates.

Photon interaction properties and water equivalence. In order to provide a first-principles comparison between the interaction properties of 4H-SiC and biological tissue, we analysed the mass attenuation coefficients (μ/ρ) of SiC and compared them to those of water, silicon, and carbon (diamond proxy), using tabulated data from the NIST XCOM database. For compound materials such as SiC, the mass attenuation coefficient can be derived under the mixture rule approximation as

$$\left(\frac{\mu}{\rho}\right)_{\text{SiC}} = w_{\text{Si}} \left(\frac{\mu}{\rho}\right)_{\text{Si}} + w_{\text{C}} \left(\frac{\mu}{\rho}\right)_{\text{C}}, \quad (1)$$

where w_{Si} and w_{C} are the elemental mass fractions and $(\mu/\rho)_{\text{Si}}$, $(\mu/\rho)_{\text{C}}$ are the corresponding mass attenuation coefficients of silicon and carbon, respectively.

Figure [2](#) reports the mass attenuation coefficients of water, SiC, silicon, and carbon (diamond proxy) over the energy range from a few keV up to 20~MeV. As expected, large deviations are observed in the low-energy region ($E \sim 50\text{keV}$), where the photoelectric effect dominates and the interaction probability scales approximately as Z^n/E^3 ($n \sim 3-4$). In this regime, silicon exhibits the strongest over-response relative to water, while SiC shows a reduced but still significant enhancement. Carbon (diamond) is closer to water due to its lower atomic number.

In the clinically relevant megavoltage range (shaded region in Figure [2](#), approximately 0.2–6 MeV), Compton scattering dominates photon interactions and the mass attenuation coefficients become primarily governed by electron density rather than atomic number. In this region, the curves for water, SiC, and Si converge significantly, indicating a substantially improved water equivalence compared to the kilovoltage domain. To better quantify deviations from water equivalence, Figure [3](#) shows the water-normalized coefficients $(\mu/\rho)_{\text{material}}/(\mu/\rho)_{\text{water}}$. In the Compton-dominated regime, SiC exhibits a nearly flat response around $\sim 0.90-1.00$, while silicon shows a slightly larger deviation at higher megavoltage energies. Above $\sim 6-8$ MeV, the increasing contribution of pair production (with

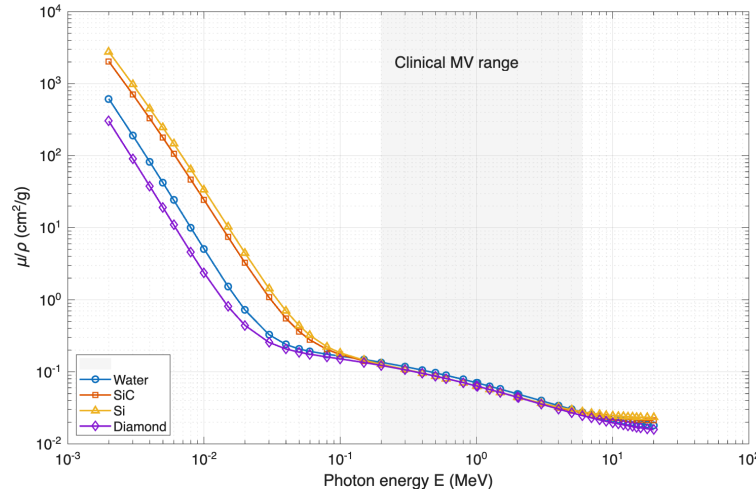


Figure 2: Mass attenuation coefficients (μ/ρ) of water, 4H-SiC, silicon, and carbon (diamond proxy) as a function of photon energy, obtained from NIST XCOM data. The shaded region indicates the clinically relevant megavoltage photon energy interval (0.2–6 MeV).

fig:MuRho

its approximate Z^2 dependence) leads to a gradual rise in the relative response of higher- Z materials. Carbon remains slightly below unity across the MV range, reflecting its lower effective atomic number. These results confirm that, although not strictly water-equivalent, 4H-SiC provides a significantly improved match to water compared to silicon in the diagnostic energy range, while maintaining near-water interaction properties in the therapeutic MV region. This theoretical analysis is consistent with the experimentally observed minimal energy dependence of the detector in 6–15 MV photon beams, as discussed in Section [II.C.](#)

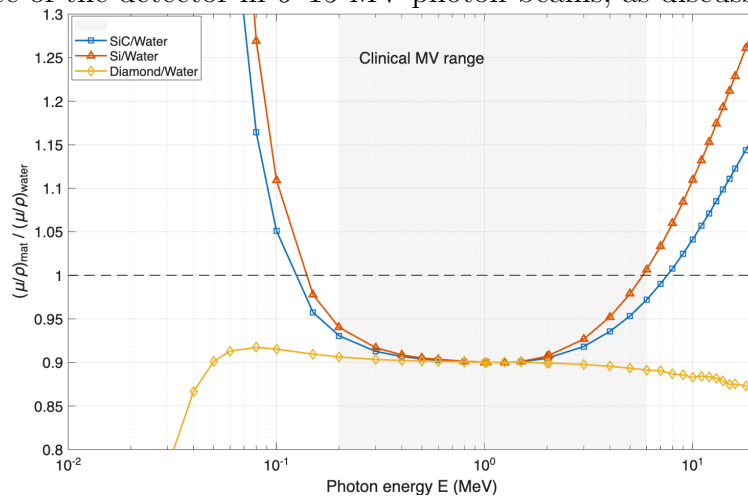


Figure 3: Water-normalized mass attenuation coefficients $(\mu/\rho)_{\text{material}}/(\mu/\rho)_{\text{water}}$ for SiC, silicon, and carbon. The dashed horizontal line represents ideal water equivalence. The shaded region highlights the megavoltage photon energy interval relevant for conventional radiotherapy.

fig:Ratio

Electrical characterization Pre-irradiation electrical characterization of the SiC dosimeter was carried out at room temperature by means of current–voltage (I–V) and capacitance–voltage (C–V) measurements, in order to assess the main properties of the p–n junction and verify its suitability for stable biased operation. The I–V characteristics were measured using a Keithley 6517B picoammeter over the bias range from -60 V to 2.5 V . The diode exhibited a pronounced rectifying behaviour, with a rectification ratio of the order of 10^5 at $\pm 2\text{ V}$, confirming the good quality of the junction. Under reverse bias, the dark current remained below 18 pA up to -60 V , consistent with the low-leakage operation expected for a wide-bandgap semiconductor. At forward voltages above approximately 2 V , the current was limited by the series resistance, estimated to be about $3\text{ k}\Omega$. Capacitance–voltage measurements were performed by applying the bias in 250 mV steps from -60 V to 2.5 V , while the capacitance was measured at a constant frequency of 1 kHz . From the measured capacitance, the depletion-layer thickness W was estimated, to first approximation, as

$$W = \frac{\varepsilon_0 \varepsilon_r A}{C}, \quad (2)$$

where $\varepsilon_r = 9.7$ is the relative dielectric constant of 4H–SiC and $A = 4\text{ mm}^2$ is the sensitive area of the device. The depletion width at zero bias was found to be approximately $1.7\text{ }\mu\text{m}$. As the reverse bias increased, the space-charge region progressively extended into the epitaxial layer, reaching about $9.5\text{ }\mu\text{m}$ at -10 V and approaching the nominal active thickness of $10\text{ }\mu\text{m}$ under higher reverse bias. These results indicate that full depletion of the epitaxial layer is achieved at approximately 10 V , while operation at -30 V corresponds to a stable over-depleted regime, consistent with the bias conditions adopted for dosimetric measurements. The measured I–V and C–V characteristics, together with the depletion-width estimate derived from the capacitance data, are reported in Figure [4](#). ^{fig:IVCV}

II.B. Experimental Setup

All measurements were carried out using a clinical Varian LINAC (Varian Clinac series, Varian Medical Systems, Palo Alto, CA, USA) delivering conventional flattened photon beams with nominal energies of 6 , 10 , and 15 MV . The accelerator is installed at the Azienda Ospedaliero Universitaria Policlinico Gaspare Rodolico in Catania and was

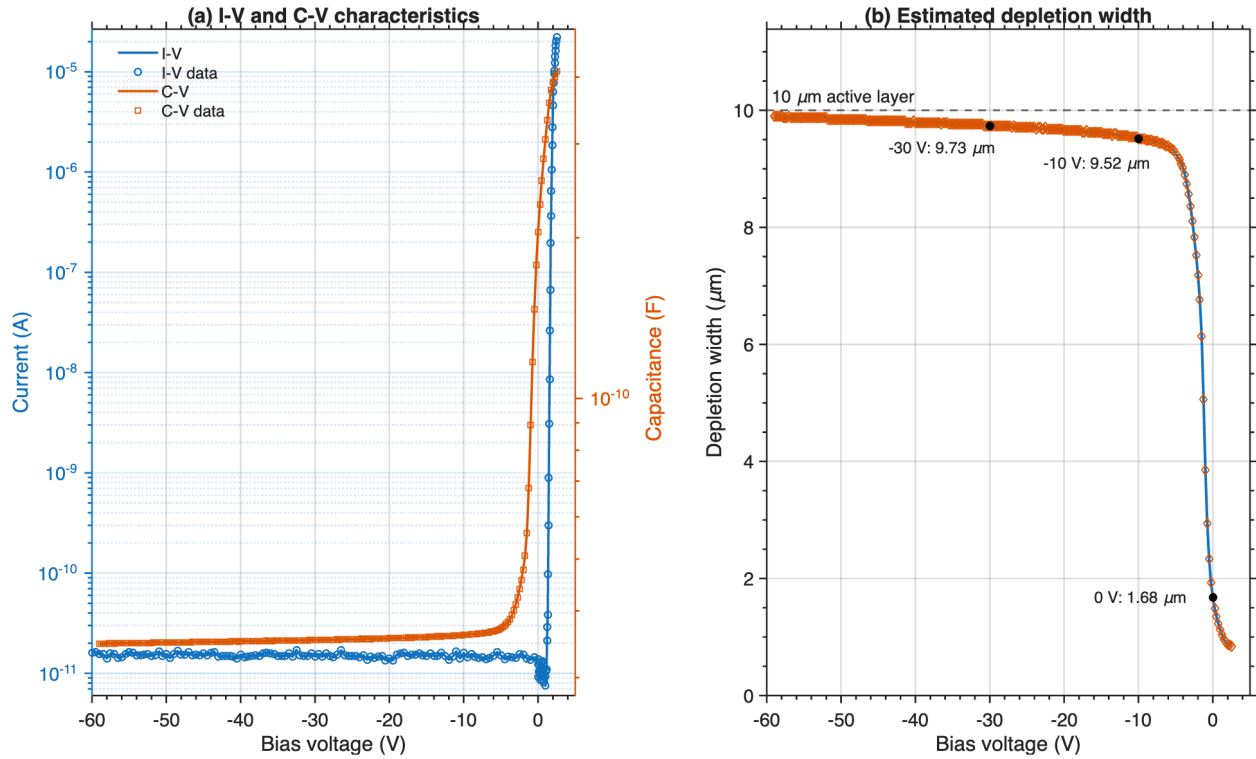


Figure 4: Pre-irradiation electrical characterization of the 4H-SiC dosimeter at room temperature. (a) Current–voltage (I–V) and capacitance–voltage (C–V) characteristics of the p–n junction shown as a function of the applied bias using a double y-axis representation; both y-axes are reported on a logarithmic scale to highlight the wide dynamic range of the electrical response. The I–V curve shows pronounced rectifying behaviour and reverse-bias dark current in the picoampere range, whereas the C–V curve reflects the progressive extension of the depletion region with increasing reverse bias. (b) Depletion-layer thickness estimated from the capacitance data. The depletion width rises rapidly at low reverse bias and approaches the nominal active-layer thickness of about 10 μm for bias values of the order of 10 V, while higher reverse bias corresponds to an over-depleted operating regime.

fig:IVCV

operated under routine clinical quality-assurance conditions, with output constancy verified before the experimental sessions. Unless otherwise stated, irradiations were performed with a source-to-surface distance (SSD) of 100 cm and a square $10 \times 10 \text{ cm}^2$ field under normal beam incidence. The gantry angle was set to 0° for standard measurements and was varied only for the dedicated angular-dependence tests. Dosimetric measurements were performed in a scanning water phantom (PTW MP3, Freiburg, Germany). The phantom was filled with water and aligned with the beam central axis under the selected irradiation geometry. The SiC detector was positioned at the required measurement depth depending on the specific test, with its active region centered on the beam axis. For depth-dose mea-

surements, the detector was mounted with its longitudinal axis parallel to the beam axis, consistently with the characterization protocol adopted for Percentage Depth Dose (PDD) acquisition. For the other measurements, the detector orientation was kept fixed unless intentionally modified for angular-response investigations. The water phantom was allowed to thermally equilibrate before data collection. The SiC device was operated under a reverse bias of 30 V and connected through a low-noise triaxial cable to a Keithley 6517B electrometer placed outside the treatment room. The electrometer was used both to supply the detector bias and to read the radiation-induced current/charge signal. Its high current sensitivity allowed stable measurement of the detector dark current as well as accurate readout of the signal generated during irradiation. A calibrated ionization-chamber reference dosimetry system was used for comparison measurements and dose verification. For PDD measurements, an Advanced Markus plane-parallel ionization chamber (PTW 34045) was used to acquire the depth-dose curve, while a PTW thimble chamber (PTW 31021) was employed as a monitor chamber to normalize the signal for possible fluctuations in beam output during the scan. Absolute output verification and reference dosimetry were performed with a PTW 30010 Farmer-type ionization chamber. For all comparative measurements, the reference detector was irradiated under the same nominal beam conditions adopted for the SiC dosimeter. The accelerator output was referenced under standard absorbed-dose-to-water calibration conditions according to the local clinical dosimetry protocol. The delivered monitor units were selected according to the purpose of each measurement, and the actual dose was verified through this reference dosimetry chain. Unless otherwise specified in the corresponding subsections, the measurements followed the characterization scheme adopted for the SiC detector, namely: dose-linearity and dose-rate tests under 6 MV irradiation; beam-quality comparison at 5 cm depth for 6 MV and 10 MV, and at 10 cm depth for 15 MV; PDD scans in water down to 25 cm depth; and field-size measurements at 5 cm depth for 6 MV and 10 cm depth for 15 MV. Ambient conditions during the experiments were monitored and remained stable throughout the measurement sessions.

II.C. Results

II.C.1. Linearity with Dose

To assess dose linearity, the SiC detector was irradiated with different delivered doses and the resulting collected charge was measured. The dose was varied by changing the number of monitor units while keeping the beam energy (6 MV), field size ($10 \times 10 \text{ cm}^2$), SSD (100 cm), and measurement depth in water (5 cm) constant. The dose per monitor unit under these conditions was known from the accelerator calibration. Doses ranging from 0.25 Gy to 10 Gy (corresponding to 10–500 MU) were delivered, and the integrated charge from the SiC detector was recorded after each irradiation. Figure 5 shows the collected charge as a function of delivered dose. The relationship was highly linear over the full investigated range. Linear regression yielded a correlation coefficient $R^2 > 0.99$, and the maximum deviation from the fit was below 1 %. No evidence of saturation or supralinear behavior was observed. The slope of the fit corresponds to the detector sensitivity; at 6 MV, a sensitivity of 92.5 nC/Gy was measured (Fig. 5b), in agreement with previous studies on SiC diode dosimeters. This excellent linearity is a key requirement for accurate dosimetry, as it enables the detector to measure both low and high delivered doses without recalibration over the investigated dynamic range.

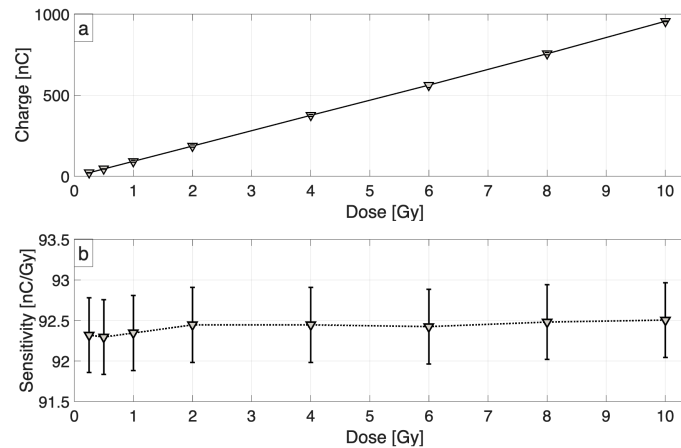


Figure 5: Linearity of the SiC detector response for 6 MV photons. (a) Integrated charge collected as a function of delivered dose (0.25 Gy to 10 Gy) under reference irradiation conditions (field size $10 \times 10 \text{ cm}^2$, SSD 100 cm, depth 5 cm in water, reverse bias 30 V). The dashed line represents the linear regression. (b) Corresponding sensitivity (charge per unit dose, nC/Gy) as a function of dose, showing deviations from constancy below the 1 % level.

fig:PlotLin

II.C.2. Dose-Rate Dependence

The dependence of the detector response on dose rate was investigated by delivering the same total dose at different LINAC output settings. The accelerator dose rate, expressed in MU/min, was varied in discrete steps from 70 MU/min to 1200 MU/min. At each setting, the irradiation time was adjusted to deliver a fixed dose of approximately 1 Gy, while keeping the total number of monitor units constant. Thus, lower dose rates corresponded to longer irradiation times, without changing the delivered dose. For each exposure, the collected charge was measured and normalized to the value obtained at the reference dose rate of 400 MU/min. A dose-rate independent response would therefore correspond to normalized values equal to unity within experimental uncertainty. As shown in Fig. 6, the normalized response remained within $\pm 1\%$ over the entire investigated range, with no evidence of a systematic trend. To reduce experimental scatter, each measurement was repeated three times and averaged; the relative standard deviation was within 0.5% for all points. No pre-irradiation history effects were observed: measurements were acquired in ascending order of dose rate, and the point at 400 MU/min was repeated at the end of the sequence to verify reproducibility.

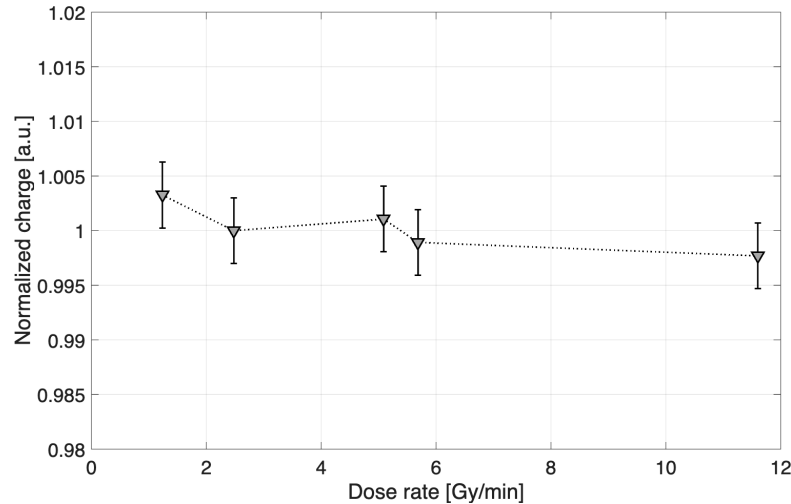


Figure 6: Dose-rate dependence of the SiC detector under 6 MV irradiation. A fixed delivered dose of 1 Gy was delivered while varying the linac dose rate from 70 MU/min to 1200 MU/min. The detector response is shown as normalized collected charge (a.u.) with respect to the value at 400 MU/min, remaining within $\pm 1\%$ over the investigated range. Error bars represent one standard deviation over repeated measurements.

fig:DoseRate

II.C.3. Energy Dependence

The detector energy response was characterized by comparing its sensitivity (collected charge per unit dose) for clinical photon beams of 6 MV, 10 MV, and 15 MV. In accordance with the measurement scheme adopted in this study, the detector was irradiated at 5 cm depth for the 6 MV and 10 MV beams and at 10 cm depth for the 15 MV beam, under otherwise identical nominal irradiation conditions (field size $10 \times 10 \text{ cm}^2$, SSD 100 cm, reverse bias 30 V). For each beam quality, a fixed dose of 2 Gy was delivered and the corresponding detector signal was recorded. The results showed that the SiC detector response was only weakly dependent on beam quality in the investigated megavoltage range. After normalization to the 6 MV response, both the 10 MV and 15 MV sensitivities remained within approximately 1 % of the baseline value. Given the combined statistical and dosimetric uncertainty, these differences are at the threshold of significance and indicate only a limited energy dependence. This behavior compares favorably with conventional silicon diodes, which may exhibit a more pronounced over-response at lower photon energies because of their higher effective atomic number. In the present case, the small residual variation may be related to the slightly higher effective atomic number of SiC relative to water, which can produce a modest over-response to the softer spectrum of the 6 MV beam. Overall, within the range from 6 to 15 MV, the detector can be regarded as energy-independent to within approximately 2 %. Figure 7 summarizes the normalized detector response for the investigated photon beam qualities.

fig:EnergyRespon

II.C.4. Percent Depth Dose Measurements

One of the most important evaluations for a new dosimeter intended for clinical beam characterization is its ability to reproduce depth–dose distributions accurately. Percent depth–dose (PDD) curves were therefore measured in water for the 6 MV, 10 MV, and 15 MV photon beams using the SiC detector and compared with reference curves acquired with an Advanced Markus plane-parallel ionization chamber (PTW 34045). During the scan, a PTW thimble chamber (PTW 31021) was used as a monitor detector to normalize the measurements for possible fluctuations in beam output. For each beam quality, the detector was positioned on the beam central axis and scanned in depth in the water phantom with its longitudinal axis kept parallel to the beam axis. The scan started near the phantom surface and extended down to 25 cm depth. Fine depth steps were used in the build-up region and around the

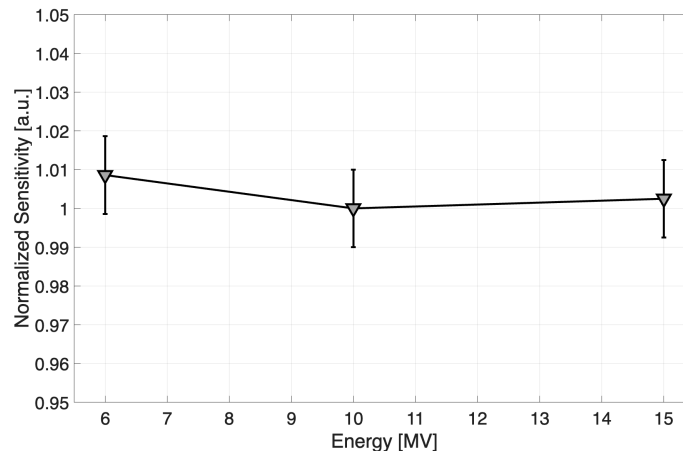


Figure 7: Energy dependence of the SiC detector sensitivity for clinical megavoltage photon beams. The response (charge per unit dose) measured at 6 MV, 10 MV, and 15 MV for 2 Gy exposures is normalized to the 6 MV value. Measurements were performed at 5 cm depth for the 6 MV and 10 MV beams and at 10 cm depth for the 15 MV beam, showing variations within approximately 2 %.

fig:EnergyR

depth of maximum dose, $d_{\max}$, whereas coarser steps were adopted in the descending portion of the curve. At each depth, a fixed number of monitor units was delivered and the corresponding SiC signal was recorded. The resulting PDD curves, shown in Fig. 8, exhibited very good agreement between the SiC detector and the reference ionization-chamber measurements for all three beam qualities. For each energy, the depth of maximum dose determined with the SiC detector matched the reference value within 1 mm. At 6 MV, for example, $d_{\max}$ was observed at approximately 15 mm depth in both measurements. In the build-up region, the SiC detector yielded slightly higher relative readings than the reference chamber. At very shallow depths, for example around 2 mm, the SiC response exceeded the reference value by about 2 %. This modest discrepancy is plausibly related to differences in detector construction and perturbation properties: the SiC device has a very small sensitive volume and minimal encapsulation, which may lead to a slightly enhanced sensitivity to low-energy scattered photons and contaminant electrons in the build-up region. Beyond $d_{\max}$, however, the SiC and reference PDD curves were essentially superimposed. The post- $d_{\max}$ fall-off and the dose values at larger depths agreed within 1 % over the investigated range. These findings demonstrate that the SiC detector can accurately reproduce clinically relevant depth-dose distributions in water for megavoltage photon beams. Although the slight over-response in the build-up region could be corrected if needed, the overall PDD agreement confirms that the SiC p-n junction is a promising solid-state detector for relative dosimetry

319 and beam characterization.

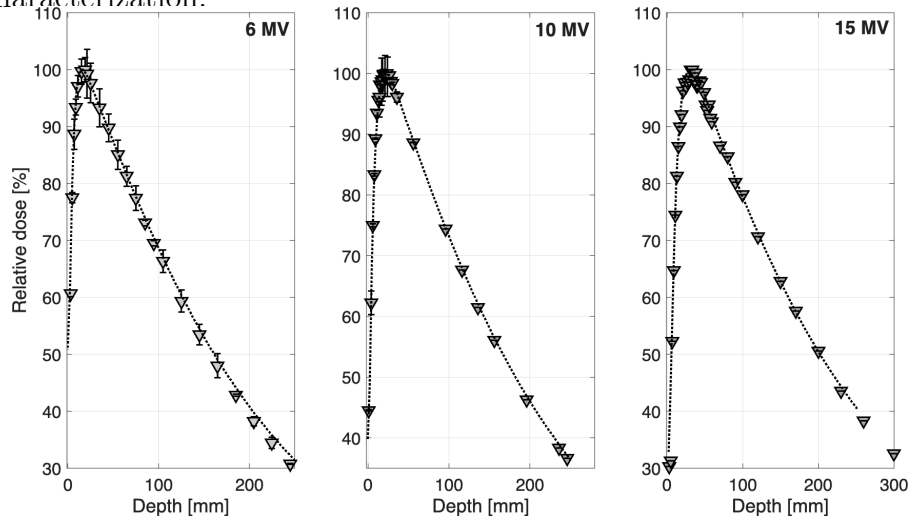


Figure 8: PDD curves in water measured with the SiC detector (symbols) and with an Advanced Markus plane-parallel ionization chamber (lines) for 6 MV, 10 MV, and 15 MV photon beams (field size $10 \times 10 \text{ cm}^2$, SSD 100 cm). The SiC detector was scanned with its axis parallel to the beam axis, and a PTW thimble chamber was used as a monitor detector to normalize the data for possible beam-output fluctuations. Curves are normalized to the maximum dose for each energy.

fig:PDD

320 II.C.5. Angular Dependence

321 The angular response of the SiC detector was investigated by varying the beam incidence
 322 direction with respect to the detector while keeping the device fixed at the linac isocenter
 323 inside the water phantom. Two independent angular scans were performed to disentangle
 324 polar and azimuthal effects. In the first configuration (θ scan), the gantry angle was varied in
 325 the vertical plane, thereby changing the polar incidence angle between the photon beam and
 326 the detector surface (Fig. [9](#), top). The reference condition ($\theta = 0^\circ$) corresponds to normal
 327 incidence, i.e. the beam perpendicular to the detector top surface. The gantry was rotated
 328 stepwise up to 90° , approaching grazing incidence (beam parallel to the phantom surface
 329 and impinging laterally on the detector). For each gantry position, the phantom alignment
 330 was carefully adjusted to keep the water-equivalent path length to the detector constant at
 331 5 cm, so that the dose at the detector position was preserved. In the second configuration
 332 (ϕ scan), the detector was rotated around its longitudinal axis while maintaining the beam
 333 direction fixed. This azimuthal rotation probes potential response asymmetries arising from
 334 the internal detector layout and encapsulation. Although the external probe has an approx-

imately cylindrical shape, the sensitive element is a planar SiC diode mounted within an asymmetric housing (epoxy resin and a metallic holder). Therefore, changing ϕ effectively alternates different irradiation conditions with respect to the diode plane and its mounting, potentially modifying the amount of non-sensitive material traversed before reaching the active layer and inducing small perturbations in build-up and attenuation. At each angular position, a low delivered dose (well below 1 Gy) was delivered and the collected charge was recorded. The results are summarized in the polar plots shown in Fig. 9, where the collected charge (in nC) is reported as a function of θ and ϕ , respectively.

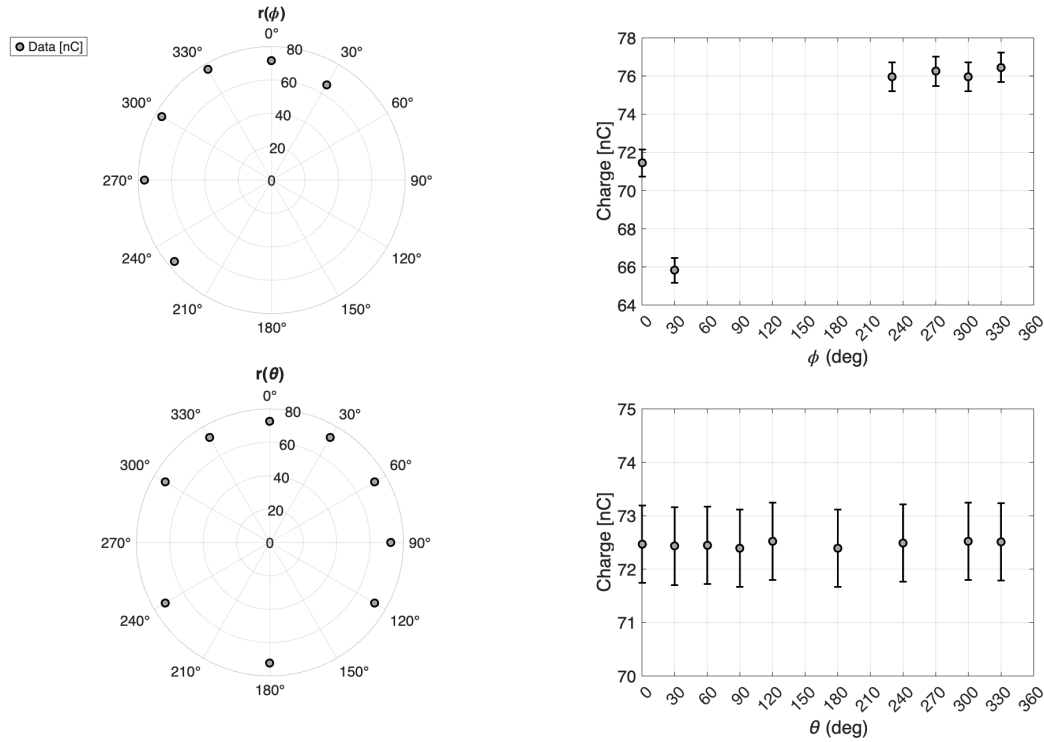


Figure 9: Angular response of the SiC detector at 5 cm depth in water. Top: polar scan (θ), obtained by varying the gantry angle in the vertical plane to change the incidence angle with respect to the detector surface while keeping the water-equivalent path length constant. Bottom: azimuthal scan (ϕ), obtained by rotating the detector around its longitudinal axis at fixed beam direction to probe potential asymmetries due to the planar diode geometry and encapsulation. In both scans, the collected charge is reported in nC.

fig:Angle

For the polar scan (θ), the response exhibited only a modest variation with respect to normal incidence, remaining within about $\sim 3\%$ over the investigated range, with no clear systematic trend up to large oblique angles (Fig. 9, top). This indicates a weak dependence on the polar incidence angle for the explored geometry. In contrast, the azimuthal scan (ϕ)

showed a pronounced modulation of the collected charge (Fig. ^{fig:Angle}9, bottom), with differences of up to a few-fold between minimum and maximum values, consistent with the intrinsic front/back asymmetry of the encapsulated planar diode within the probe. Practically, these findings indicate that the detector orientation around its axis should be controlled (or appropriate correction factors should be applied) whenever non-coplanar or lateral irradiation geometries are expected.

II.C.6. Temperature Dependence

Lastly, the influence of temperature on the detector response was investigated under controlled conditions. The water-phantom temperature was varied between 20 °C and 40 °C and monitored with a calibrated thermometer. At each temperature point, the detector, positioned at 5 cm depth under a 6 MV photon beam and SSD = 100 cm, was irradiated with a fixed dose of 1 Gy, and the corresponding signal was recorded. The reference ionization chamber was used to verify the stability of the delivered dose throughout the measurements. A slight decrease in the SiC signal was observed with increasing temperature. Between 20 °C and 40 °C, the total variation in response was approximately 3 %, corresponding to an average temperature coefficient of about $-0.2\% \text{ } ^\circ\text{C}^{-1}$. The negative sign indicates a reduction in sensitivity with increasing temperature. Figure ^{fig:Temperature}10 illustrates this trend. The magnitude of this temperature dependence is relatively small. For comparison, silicon diode dosimeters often exhibit larger temperature coefficients, typically in the range from about $-0.3\% \text{ } ^\circ\text{C}^{-1}$ to $-0.5\% \text{ } ^\circ\text{C}^{-1}$, which may require explicit correction during clinical use. The SiC detector therefore shows a comparatively stable response over the investigated temperature range, closer to that typically observed for diamond-based dosimeters.

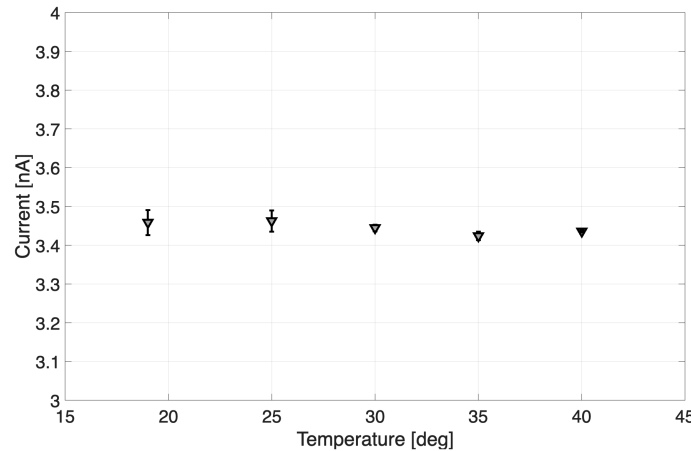


Figure 10: Temperature dependence of the SiC detector response. The detector current measured under 6 MV irradiation at 5 cm depth in water (SSD 100 cm) for a fixed delivered dose of 1 Gy is reported as a function of phantom temperature (20 °C to 40 °C). A small negative temperature trend is observed over the investigated range.

fig:Temperat

III. Conclusions

This study presented a dosimetric characterization of a novel 4H-SiC planar p-n junction detector operated in clinical megavoltage photon beams. Measurements were carried out in a water phantom under reference conditions and benchmarked against an absorbed-dose-to-water calibrated ionization chamber, following a formalism consistent with international and AAPM recommendations for MV photon reference dosimetry [22,23,24]. The main goal was to verify whether the SiC material properties and device design can provide a stable, high signal-to-noise response suitable for routine clinical measurements, including depth-dose verification and QA-oriented checks. From an electrical standpoint, the detector showed clear diode rectification, ultra-low reverse leakage current at the operating bias, and depletion behavior compatible with full depletion at voltages well below the applied reverse bias. Operation in an over-depleted regime supports saturated carrier drift and near-complete charge collection in the active region, providing a solid basis for current-mode readout with limited noise and without stringent thermal requirements. Dosimetrically, the device fulfilled the main criteria expected for relative MV photon dosimetry. The collected charge increased linearly with delivered dose over the investigated interval, with sub-percent deviations from linearity. The response was stable across clinically relevant dose-rate settings, remaining constant within experimental uncertainty over the explored MU/min range. Beam-quality dependence was minimal between 6 and 15 MV: after normalization, sensitivity changes stayed within a few

percent, indicating that the effective atomic number of SiC does not introduce clinically relevant energy dependence in this range. This behaviour compares favorably with conventional silicon diodes and supports the role of SiC as a competitive solid-state option when near water-equivalent response and manufacturability are both desired ¹⁹. The detector also reproduced percent depth-dose curves in water for 6, 10 and 15 MV beams, with agreement on $d_{\max}$ within typical experimental positioning uncertainties and percent-level agreement in the descending region. A modest over-response was observed in the build-up region at shallow depths, plausibly linked to spectral effects (enhanced sensitivity to low-energy components and contaminant electrons) and to packaging-related perturbations. Angular dependence showed a limited polar effect (within a few percent over large incidence angles), while a pronounced azimuthal modulation was observed, attributable to the non-symmetric encapsulation/holder rather than to a fundamental limitation of SiC. In practice, this can be mitigated by enforcing a fixed azimuthal orientation during use and/or by redesigning the packaging toward a more symmetric geometry. Temperature tests indicated a small negative temperature coefficient over the explored range, which is not expected to be limiting under typical water-phantom conditions but may be accounted for through simple monitoring or correction in more demanding scenarios. Overall, the proposed SiC p-n junction detector emerges as a strong candidate for relative dosimetry and QA measurements in MV photon beams, particularly where high spatial resolution, real-time readout, and operational stability are beneficial. Future work should extend the validation to small-field and stereotactic conditions using established codes of practice, including appropriate detector-specific correction strategies and orientation control ²⁰. The main residual limitations identified here are primarily engineering-driven (build-up behaviour and azimuthal angular dependence) and therefore addressable, supporting continued development toward optimized, clinically deployable, protocol-traceable SiC dosimetry solutions.

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